

```
import os

if not os.path.exists('SMSSpamCollection'):
    !wget -q "https://archive.ics.uci.edu/ml/machine-learning-databases/00228/smsspamcollection.zip"
    !unzip -q smsspamcollection.zip
    print("✅ Dataset downloaded successfully!")
else:
    print("✅ Dataset already exists, skipping download!")
```

✅ Dataset already exists, skipping download!

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.svm import LinearSVC
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

print("✅ All libraries imported!")
```

✅ All libraries imported!

```
# Load the dataset
df = pd.read_csv('SMSSpamCollection', sep='\t', header=None, names=['label', 'message'])

print("Shape:", df.shape)
print("\nFirst 5 rows:")
df.head()
```

Shape: (5572, 2)

First 5 rows:

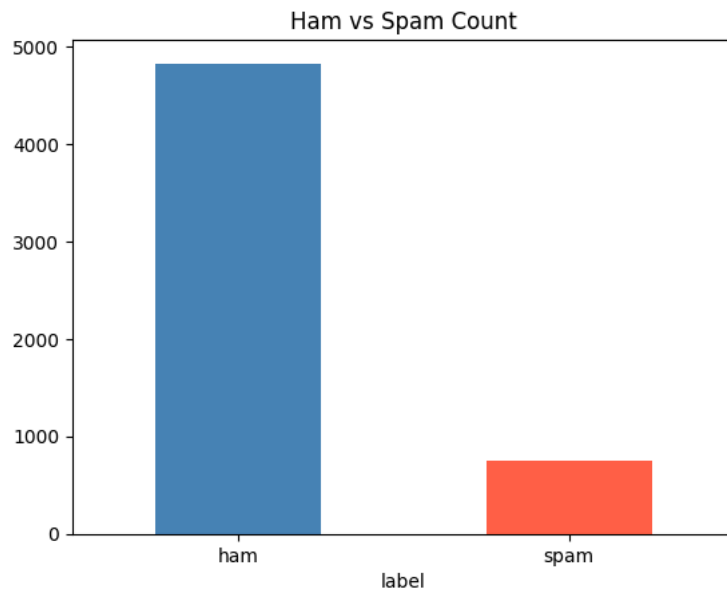
	label	message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
# Check how many spam vs ham
print(df['label'].value_counts())

# Visualize
df['label'].value_counts().plot(kind='bar', color=['steelblue', 'tomato'])
plt.title('Ham vs Spam Count')
plt.xticks(rotation=0)
plt.show()
```

```
label
ham      4825
spam      747
Name: count, dtype: int64
```



```
# Convert labels to numbers: ham=0, spam=1
df['label_num'] = df['label'].map({'ham': 0, 'spam': 1})

# Add message length as a feature (optional but insightful)
df['msg_length'] = df['message'].apply(len)

print("✅ Preprocessing done!")
df.head()
```

✅ Preprocessing done!

	label	message	label_num	msg_length	
0	ham	Go until jurong point, crazy.. Available only ...	0	111	
1	ham	Ok lar... Joking wif u oni...	0	29	
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	1	155	
3	ham	U dun say so early hor... U c already then say...	0	49	
4	ham	Nah I don't think he goes to usf, he lives aro...	0	61	

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
# Split data
X = df['message']
y = df['label_num']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Convert text to numbers using TF-IDF
vectorizer = TfidfVectorizer(stop_words='english', max_features=5000)
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)

# Train Naive Bayes
nb_model = MultinomialNB()
nb_model.fit(X_train_vec, y_train)
nb_preds = nb_model.predict(X_test_vec)

# Train SVM
svm_model = LinearSVC()
svm_model.fit(X_train_vec, y_train)
svm_preds = svm_model.predict(X_test_vec)

print("✅ Models trained!")
```

✅ Models trained!

```

print("=" * 40)
print("NAIVE BAYES RESULTS")
print("=" * 40)
print(f"Accuracy: {accuracy_score(y_test, nb_preds)*100:.2f}%")
print(classification_report(y_test, nb_preds, target_names=['Ham', 'Spam']))

print("=" * 40)
print("SVM RESULTS")
print("=" * 40)
print(f"Accuracy: {accuracy_score(y_test, svm_preds)*100:.2f}%")
print(classification_report(y_test, svm_preds, target_names=['Ham', 'Spam']))

```

NAIVE BAYES RESULTS

```

=====
Accuracy: 98.21%

```

	precision	recall	f1-score	support
Ham	0.98	1.00	0.99	966
Spam	0.99	0.87	0.93	149
accuracy			0.98	1115
macro avg	0.99	0.94	0.96	1115
weighted avg	0.98	0.98	0.98	1115

```

=====
SVM RESULTS
=====
Accuracy: 98.92%

```

	precision	recall	f1-score	support
Ham	0.99	1.00	0.99	966
Spam	0.99	0.93	0.96	149
accuracy			0.99	1115
macro avg	0.99	0.96	0.98	1115
weighted avg	0.99	0.99	0.99	1115

```

def predict_spam(message):
    msg_vec = vectorizer.transform([message])
    nb_result = nb_model.predict(msg_vec)[0]
    svm_result = svm_model.predict(msg_vec)[0]
    print(f"Message: '{message}'")
    print(f" Naive Bayes → {'🔥 SPAM' if nb_result == 1 else '✅ HAM (Not Spam)'}")
    print(f" SVM → {'🔥 SPAM' if svm_result == 1 else '✅ HAM (Not Spam)'}")
    print()

# Test it!
predict_spam("Congratulations! You've won a FREE iPhone. Click here now!")
predict_spam("Hey, are we still meeting for lunch tomorrow?")
predict_spam("URGENT: Your bank account has been compromised. Call now!")

```

```

Message: 'Congratulations! You've won a FREE iPhone. Click here now!'
Naive Bayes → 🔥 SPAM
SVM → 🔥 SPAM

Message: 'Hey, are we still meeting for lunch tomorrow?'
Naive Bayes → ✅ HAM (Not Spam)
SVM → ✅ HAM (Not Spam)

Message: 'URGENT: Your bank account has been compromised. Call now!'
Naive Bayes → ✅ HAM (Not Spam)
SVM → ✅ HAM (Not Spam)

```